Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities Minneapolis, MN
*Bachelor of Science in **Mathematics*** *Expected May 2028*

- **GPA:** 4.0/4.0 **| Honors:** Dean's List, Fall 2025, Spring 2026
- **Relevant Coursework:** Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists and Engineers I & II (A), Introduction to Programming Concepts (Python, A), Introduction to Statistical Analysis (A)

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota Minneapolis, MN
Undergraduate Research Assistant & UROP Fellow, Advisor: Prof. Yan Liu Oct 2025 - Present

- Built a **Python data pipeline** (Oct 2025) in Jupyter Notebook to process LED calibration data from 11 cryogenic germanium detector channels: implemented baseline subtraction, peak detection, time alignment, and exponential fitting; processed 10+ GB of raw data using NumPy/Pandas with QA visualizations.
- Collaborated with postdoctoral researcher Shubham Pandey on biweekly progress reviews; integrated template-matching methods into the experiment's reconstruction workflow; conducted data analysis and project development on the MSI Agate HPC cluster
- Participated in the physical **installation of a BlueFors dilution refrigerator** (May 18–20, 2026) under guidance of a BlueFors field engineer; independently set up the compressor water-cooling circulation; assembled a heater on the 4K plate to diagnose elevated temperature and identified high P5 valve pressure as the root cause; assisted with component assembly, vacuum pump-down, and initial 4K cooldown verification.
- Extended RuOx thermometer **calibration curves** (sensors R31279 & R31839) from factory range to the full 5 mK–300 K operational range (May 23–27, 2026); implemented three fitting models (degree-5 log-polynomial, PCHIP + Mott VRH, cubic log-polynomial) and generated validated Lake Shore .340 files for direct controller loading.

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow Beijing, China
Team Leader Oct 2024 - Jun 2025

- Led a student team to design a cost-efficient billiards rest prototype addressing equipment inefficiencies;
- Conducted **market analysis and product testing** to evaluate commercialization potential;
- Invited to present the project at **Tsinghua X-Lab & Kering Entrepreneurship Roadshow**, communicating with a high-profile AI team and presenting to industry leaders.

Low-Carbon Transition Project - China Steel Industry Beijing, China
Undergraduate Researcher Oct 2024 - May 2025

- Assisted in research on low-carbon transition pathways for China's steel industry;
- Contributed to **data collection, database construction, and quantitative analysis** for national low-carbon initiatives.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **First Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **First Prize**, National College E-Commerce "Innovation, Creativity, and Entrepreneurship" Challenge, Campus Level | Team Leader
- **Third Prize**, China International College Students' Innovation Competition, Campus Level (2025) | Team Leader
- **Second Prize**, NECCS National English Competition for College Students

ACTIVITIES & INVOLVEMENT

- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, waveform template construction, QA visualization
- **Tools:** Python, Java, NumPy, Pandas, Matplotlib, SciPy, Git/GitHub, Jupyter Notebook, MSI HPC (Agate), Excel
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)